

Software Engineering

LAB 1

Requirements Engineering & UML Use-Case Modelling

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Problem statement #09:

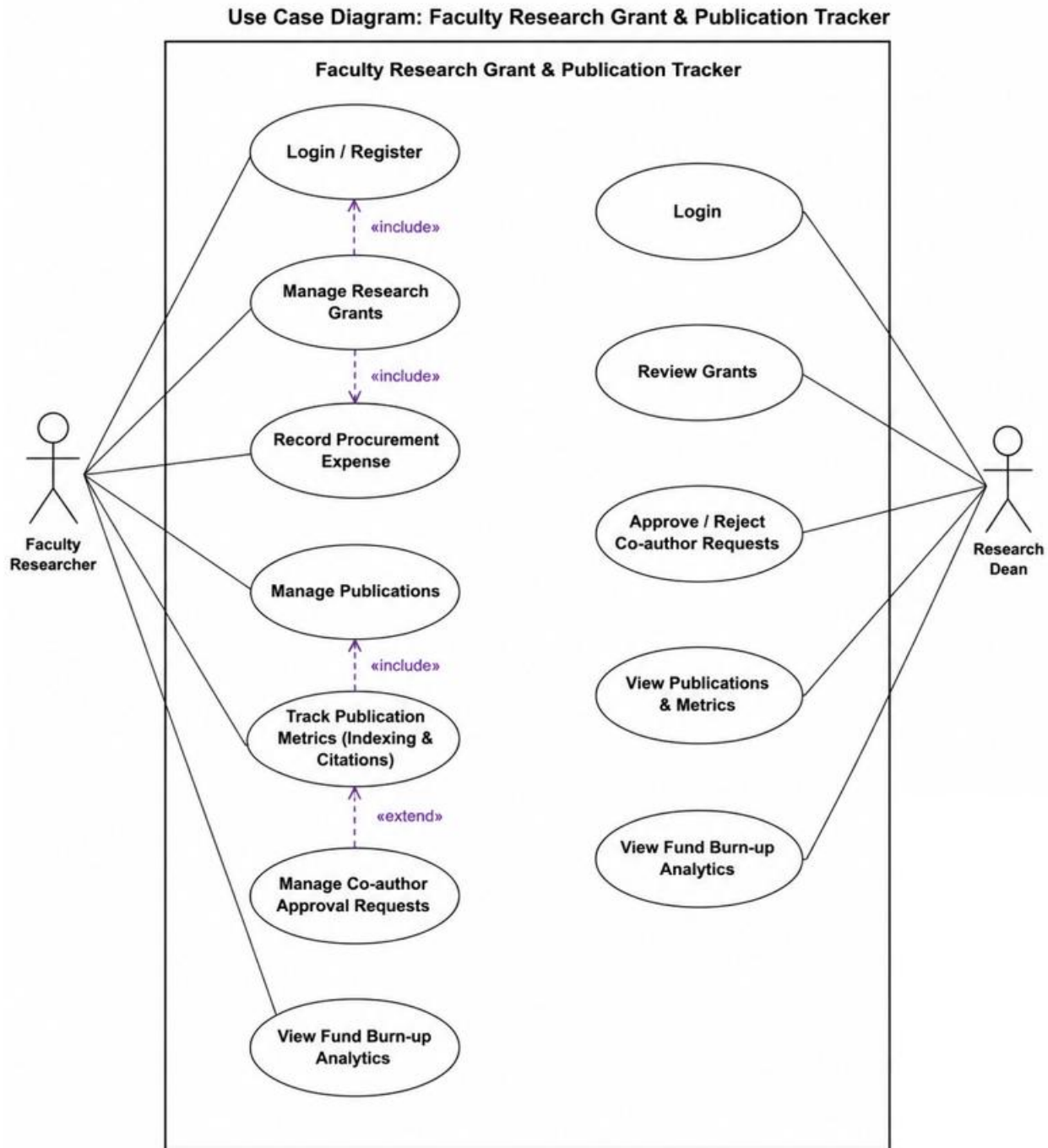
Faculty Research Grant & Publication Tracker

Target Stakeholders / Actors: Faculty Researcher, Research Dean

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute the remaining budgetary balance.	High	Pass: Balance updates accurately and expenses exceeding the grant cap are blocked. Fail: Overdraft is allowed without admin override.	Ensures accurate tracking and control of sponsored research grant funds.
FR-002	Functional	The system shall allow the Faculty Researcher to record and update publication details, including indexing status and citation information.	High	Pass: A publication is saved with its required details and its indexing/citation information is displayed. Fail: A publication cannot be saved with incomplete required information.	Maintains accurate faculty publication records and metrics.
FR-003	Functional	The system shall allow the Faculty Researcher to submit co-author approval requests for publications and track their approval status.	High	Pass: A submitted request is recorded with a status of Pending, Approved, or Rejected. Fail: The request status cannot be tracked.	Provides a controlled workflow for co-author approvals.
FR-004	Functional	The system shall allow the Research Dean to review and approve or reject co-author approval requests.	High	Pass: The Research Dean can select a pending request and change its status to Approved or Rejected. Fail: A pending request cannot be reviewed or updated.	Enables the Research Dean to control the approval workflow.
FR-005	Functional	The system shall generate fund burn-up analytics showing the allocated, spent, and remaining funds for sponsored research grants.	High	Pass: The displayed allocated, spent, and remaining amounts match the recorded grant transactions. Fail: The displayed values do not match the recorded transactions.	Helps the Research Dean monitor research grant fund utilization.
NFR-001	Nonfunctional – Performance & Security	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Audit records cannot be modified or deleted through normal system operations and modifications are traceable. Fail: Existing audit records can be altered without detection.	Protects the integrity and traceability of financial and journal-related activities.
NFR-002	Nonfunctional – Performance	The system shall generate fund burn-up analytics within 3 seconds for a faculty researcher's grant records.	Medium	Pass: Analytics are displayed within 3 seconds during testing. Fail: Analytics take more than 3 seconds.	Ensures timely access to grant fund information.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

- Use Case Name: Manage Research Grant Funds
- Primary Actor: Faculty Researcher
- Supporting Actor: Research Dean

Preconditions

1. The Faculty Researcher has access to the research grant management system.
2. The selected sponsored research grant exists in the system.
3. The grant has an approved fund allocation.
4. The Faculty Researcher has permission to record procurement expenses.

Postconditions

1. The valid procurement expense is recorded against the selected grant.
2. The remaining grant balance is recalculated.
3. The updated grant balance is displayed to the Faculty Researcher.
4. The financial modification is recorded in the audit ledger.

Main Success Scenario

1. Faculty Researcher selects **Manage Research Grants**.
2. System displays the Faculty Researcher's available sponsored research grants.
3. Faculty Researcher selects a research grant.
4. System displays the grant allocation, total spent amount, and remaining balance.
5. Faculty Researcher selects **Record Procurement Expense**.
6. Faculty Researcher enters the equipment procurement details and expense amount.
7. System validates the entered expense information.
8. System checks whether the expense is within the remaining grant balance.
9. System accepts the expense because sufficient funds are available.
10. System records the equipment procurement expense.
11. System recalculates the remaining grant balance.
12. System records the financial modification in the audit ledger.
13. System displays the updated grant balance.
14. Use case ends successfully.

Alternate Flow 1 — Grant Limit Exceeded

8a. Expense exceeds remaining grant balance

8a1. System detects that the entered expense exceeds the available grant balance.

8a2. System rejects the expense.

8a3. System displays: "Expense exceeds the available grant balance."

8a4. Faculty Researcher is prompted to enter a lower expense amount or cancel the transaction.

8a5. No invalid financial transaction is recorded.

Alternate Flow 2 — Invalid Expense Details

7a. Invalid or incomplete expense information

7a1. System detects missing or invalid procurement expense information.

7a2. System displays the fields requiring correction.

7a3. Faculty Researcher corrects the information.

7a4. System validates the corrected information.